复旦数学资料汇编
代数几何进阶

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第一章　代数几何进阶

1.1　试卷: 代数几何进阶

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1.2　2025 秋季 (周杨)

第一题（共 30 分）

We work over an algebraically closed field. Let C be a connected smooth projective curve of genus g. Let $f : C \to \mathbb{P}^n$ be an embedding of degree d, thus $f_*([C])$ is d times the generator of $A_1(\mathbb{P}^n)$, where $[C]$ is the fundamental class of C.

1. Compute the degree of f_*N, where N is the normal bundle of $f(C)$.
2. Let $\pi : X \to \mathbb{P}^n$ be the blow up along $f(C)$ and let E be the exceptional divisor. Compute $\int_X E^n$.

Solutions

We identify C with its image $f(C)$.

1. The degree of a vector bundle over a proper curve is the degree of its first Chern class, which is additive for short exact sequences. We have a short exact sequence

$$0 \to T_C \to f^*T_{\mathbb{P}^n} \to N \to 0.$$

Taking first Chern classes we have

$$c_1(f^*T_{\mathbb{P}^n}) = c_1(T_C) + c_1(N).$$

Note that $c_1(f^*T_{\mathbb{P}^n}) = (n+1)H$, where H is the class of a hyperplane. Hence by the projection formula

$$\int_C c_1(f^*T_{\mathbb{P}^n}) = \int_{\mathbb{P}^n} (n+1)H \cdot f_*([C]) = (n+1)d.$$

Moreover,

$$\int_C c_1(T_C) = 2 - 2g.$$

Hence

$$\deg(N) = (n+1)d - 2 + 2g.$$

2. Note that $E \cong \mathbb{P}(N)$ with projection $\pi : \mathbb{P}(N) \to C$, and $\mathcal{O}(E)|_E$ is the relative $\mathcal{O}_{\mathbb{P}(N)}(-1)$. Let $\xi = c_1(\mathcal{O}_{\mathbb{P}(N)}(1))$, then we have

$$\xi^{n-1} + \pi^* c_1(N) \cdot \xi^{n-2} = 0,$$

since $c_2(N) = c_3(N) = \cdots = 0$. We have

$$\begin{aligned}\int_X E^n &= \int_X \big(c_1(\mathcal{O}(E))\big)^{n-1} \cdot E \\ &= \int_E \big(c_1(\mathcal{O}(E)|_E)\big)^{n-1} \\ &= \int_E (-1)^n \, \pi^*(c_1(N)) \cdot \xi^{n-2}.\end{aligned}$$

Note that $\pi_*(\xi^{n-2}) = [C]$. Hence by the projection formula

$$\int_X E^n = \int_C (-1)^n c_1(N) = (-1)^n \big((n+1)d - 2 + 2g\big).$$

第二题（共 30 分）

We work over an algebraically closed field. Let $(x_0, \ldots, x_3)$ be the homogeneous coordinates on $\mathbb{P}^3$. Let

- $p \in \mathbb{P}^3$ be a closed point;
- $H_i \subset \mathbb{P}^3$ be the line defined by $x_i = 0$, $i = 0, \ldots, 3$;
- $Z = H_0 \cup H_1 \cup H_2 \cup H_3$.

Let $\iota_p, \iota_{H_i}, \iota_Z$ be their inclusion morphisms into $\mathbb{P}^3$.

1. Compute $\mathrm{ch}(\iota_{p,*}(\mathcal{O}_p))$.
2. Compute $\mathrm{ch}(\iota_{H_0,*}(\mathcal{O}_{H_0}))$.
3. Compute $\mathrm{ch}(\iota_{Z,*}(\mathcal{O}_Z))$.

Solutions

1. By the Grothendieck–Riemann–Roch theorem, let N_p be the normal "bundle" of p, then

$$\mathrm{ch}(\iota_{p,*}(\mathcal{O}_p)) = \iota_{p,*}\big(\mathrm{ch}(\mathcal{O}_p) \cdot \mathrm{Td}(N_p)^{-1}\big) = \iota_{p,*}(1 \cdot 1) = [p].$$

2. Let H be the class of a hyperplane. Then

$$\mathrm{ch}(\mathcal{O}_{\mathbb{P}^3}(-1)) = e^{-H}.$$

We have a resolution

$$0 \to \mathcal{O}_{\mathbb{P}^3}(-1) \to \mathcal{O}_{\mathbb{P}^3} \to \iota_{H_0,*}(\mathcal{O}_{H_0}) \to 0.$$

Thus in K-theory,

$$\iota_{H_0,*}(\mathcal{O}_{H_0}) = \mathcal{O}_{\mathbb{P}^3} - \mathcal{O}_{\mathbb{P}^3}(-1).$$

Since $\mathrm{ch}(\cdot)$ is additive for K-theory classes,

$$\mathrm{ch}(\iota_{H_0,*}(\mathcal{O}_{H_0})) = 1 - e^{-H}.$$

3. Similarly, from the short exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^3}(-4) \to \mathcal{O}_{\mathbb{P}^3} \to \iota_{Z,*}(\mathcal{O}_Z) \to 0,$$

we have

$$\mathrm{ch}(\iota_{Z,*}(\mathcal{O}_Z)) = 1 - e^{-4H}.$$

第三题（共 40 分）

We work over an algebraically closed field k. Consider the following moduli stack. For a scheme S, an object ξ over S is a pair

$$\xi = (E, \phi)$$

where E is a vector bundle of rank 2 over S and

$$\phi = (\phi_1, \dots, \phi_4) \in H^0(S, E^{\oplus 4})$$

such whenever we have a local trivialization $\sigma : E|_U \simeq \mathcal{O}_U^{\oplus 2}$, all the 2×2 minors of the 2×4 matrix $\sigma(\phi)$ (with entries in $H^0(U, \mathcal{O}_U)$) vanish.

Given another object $\xi' = (E', \phi')$ over S' and a morphism $f : S' \to S$, an arrow $\theta : \xi' \to \xi$ lifting f is a pair $\theta = (f, g)$, where $g : E' \to f^*E$ is an isomorphism of vector bundles such that $g(\phi') = f^*\phi$.

Now fix $\xi^\circ = (k^2, \vec{0} = (0,0,0,0))$ over $\operatorname{Spec} k$. In this problem we study the deformations of ξ°. Let $\mathcal{C}$ be the category of Artin local k-algebras.

1. Write down one precise definition of the (not-crude) functor $F : \mathcal{C} \to$ Set of deformations of ξ°.
2. Compute the (dimension of) the Zariski tangent space t_F.
3. Find an obstruction theory for the moduli problem and find some deformation of ξ° whose higher order deformation is obstructed.
4. Determine if the functor has a versal/miniversal/universal deformation. And briefly explain.

Solutions

For any ring R, we denote by $R^{m\times n}$ the set of all $m \times n$ matrices with entries in R.

1. For $(A, \mathfrak{m}_A) \in \mathcal{C}$, the set $F(A)$ is

$$\left\{ X \in A^{2\times 4} \;\middle|\; \text{all } 2\times 2 \text{ minors of } X \text{ vanish, and } X \equiv 0 \pmod{\mathfrak{m}_A} \right\} \Big/ \sim,$$

where $X \sim X'$ iff there exists $T \in A^{2\times 2}$ such that

$$T \equiv I_2 \pmod{\mathfrak{m}_A} \quad \text{and} \quad TX = X',$$

and I_2 is the identity matrix of size 2.

For a map $A \to A'$, the map $F(A) \to F(A')$ is the obvious induced map.

Or equivalently, an element of $F(A)$ is represented by (M, ϕ, κ), where M is a free A-module of rank 2, $\phi \in M^{\oplus 4}$ satisfies the vanishing-minors-condition for any A-isomorphism $M \simeq A^{\oplus 2}$, and

$\kappa : M \otimes_A k \to k^2$ is a k-linear isomorphism. Two such triples (M, ϕ, κ) and (M', ϕ', κ') represent the same element iff there exists an isomorphism $\eta : M \to M'$ of A-modules making the diagram commute:

$$\begin{array}{ccc} M \otimes_A k & \xrightarrow{\eta \otimes 1_k} & M' \otimes_A k \\ {\scriptstyle\kappa}\downarrow & & \downarrow{\scriptstyle\kappa'} \\ k^2 & = & k^2 \end{array}$$

2. Let $D = k[\epsilon]$ with $\epsilon^2 = 0$ be the dual numbers. Using the first description above, an element of $F(D)$ is uniquely represented by

$$X = \epsilon Y,$$

where $Y \in k^{2\times 4}$ is arbitrary. It is straightforward to check that the k-linear structure is just the standard one on $k^{2\times 4}$. Hence

$$\dim t_F = 8.$$

3. Let

$$0 \to J \to A' \to A \to 0$$

be an extension of the Artin local k-algebra $(A, \mathfrak{m}_A)$ such that $\mathfrak{m}_A J = 0$. Given an object over A represented by $X = (a_{ij}) \in A^{2\times 4}$, pick any lift $X' = (a'_{ij}) \in (A')^{2\times 4}$. Then the 6-tuple

$$\left(\det \begin{pmatrix} a'_{1i} & a'_{1j} \\ a'_{2i} & a'_{2j} \end{pmatrix} \right)_{1 \le i < j \le 4} \in J^{\oplus 6}$$

is an element in $J^{\oplus 6}$. Different choices of the lift give the same element, since the partial derivatives of the 2×2-minors (as polynomials in the entries) vanish at the zero matrix. This defines the obstruction class, whose vanishing is equivalent to the existence of a lift to an object over A'. Naturality is obvious, thus $k^{\oplus 6}$ together with this assignment is an obstruction theory.

To find an obstructed deformation, let $A = k[t]/(t^2)$ and $A' = k[t]/(t^3)$. Take

$$X = \begin{pmatrix} t & 0 & 0 & 0 \\ 0 & t & 0 & 0 \end{pmatrix}.$$

Then the obstruction class is the class of $(t^2, 0, \ldots, 0)$ in $\big((t^2)/(t^3)\big)^{\oplus 6}$.

4. It has a miniversal deformation but no universal deformation.

One approach is to write down a miniversal deformation directly. The base of the miniversal deformation is

$$R = k\big[\big[\{x_{ij}\}_{1\le i\le 2,\, 1\le j\le 4}\big]\big] \Big/ \big(\text{all the } 2 \times 2\text{-minors}\big),$$

and the universal family is given by (the formal completion of) the "tautological" matrix

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \end{pmatrix}.$$

- We verify that the map $h_R \to F$ induced by the universal family is surjective. Given any $X_A \in A^{2\times 4}$ representing an element in $F(A)$, write $X_A = (a_{ij})$. Then along the map $R \to A$ sending $x_{ij} \mapsto a_{ij}$, the universal family pulls back to the given one.

- We verify that it is strongly surjective. Given an extension

$$0 \to J \to A' \to A \to 0$$

as above, given
  - a family $X_A \in A^{2\times 4}$,
  - a family $X_{A'} \in (A')^{2\times 4}$,
  - a map $\varphi : R \to A$,
  - a matrix $T \in A^{2\times 2}$, $T \equiv I_2$,

such that $\varphi(X) = TX_A$ and $X_{A'} \equiv X_A \pmod J$. We need to lift φ to $\varphi' : R \to A'$ and find a matrix $T' \in (A')^{2\times 2}$ such that $T' \equiv I_2$ and $\varphi'(X) = T'X_{A'}$. To do this, find any lift T' of T and define φ' by sending x_{ij} to the (i,j)-entry of $T'X_{A'}$.
- It is immediate that $h_R(D) \to F(D)$ is an isomorphism.

It does not have a universal family. A miniversal family is unique up to (possibly non-unique) isomorphism. Hence it suffices to show that the miniversal family is not universal. To see this, take $A = k[t]/(t^3)$, then

$$\begin{pmatrix} 1+t & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} t+t^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} t+2t^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Hence, the two maps $R \to A$ sending x_{11} to $t+t^2$ and to $t+2t^2$ (and other x_{ij} to 0) induce the same element in $F(A)$. Hence it is not universal.

Another approach is to use Schlessinger's criterion. As in many deformation problems, given an object ξ over A the set

$$\frac{\{\xi' \mid \xi' \text{ is a lift of } \xi \text{ to an object over } A'\}}{\{\text{isomorphisms of } \xi' \text{ which is identity on } \xi\}}$$

is a pseudo-torsor over t_F. Thus one can apply Schlessinger's criterion to show that a miniversal family exists. To show that it is not universal, it suffices to show that there is an automorphism of ξ that

- restricts to identity over $k = A/\mathfrak{m}_A$, and
- does not lift to an automorphism of ξ'.

Let $A = k[t]/(t^2)$ and $A' = k[t]/(t^3)$. Let ξ be the object represented by

$$\begin{pmatrix} t & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

An automorphism of ξ restricting to identity on k is $I_2 + tB$, for arbitrary $B \in k^{2\times 2}$. However, if we take $B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, then it does not lift to an automorphism of the object ξ' represented by

$$X_{A'} = \begin{pmatrix} t+t^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

since for any $C \in k^{2\times 2}$, we always have

$$(I + tB + t^2C)X_{A'} = \begin{pmatrix} t+2t^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \neq X_{A'}.$$

[Note:] We are actually describing the stack $[Z/GL(2,k)]$, where $Z \subset \mathbb{A}^{2\times 4}$ is the space of 2×4 matrices with rank at most 1. For more on formal moduli in the context of stacks, see Alper's *Stacks and Moduli*.